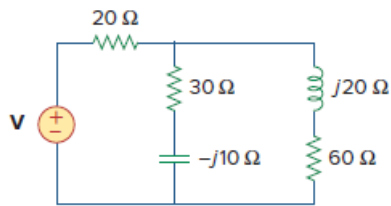


Problem

In the circuit in Fig. 11.25, the $60\text{-}\Omega$ resistor absorbs an average power of 240 W . Find $\mathbf{V}$ and the complex power of each branch of the circuit. What is the overall complex power of the circuit? (Assume the current through the $60\text{-}\Omega$ resistor has no phase shift.)

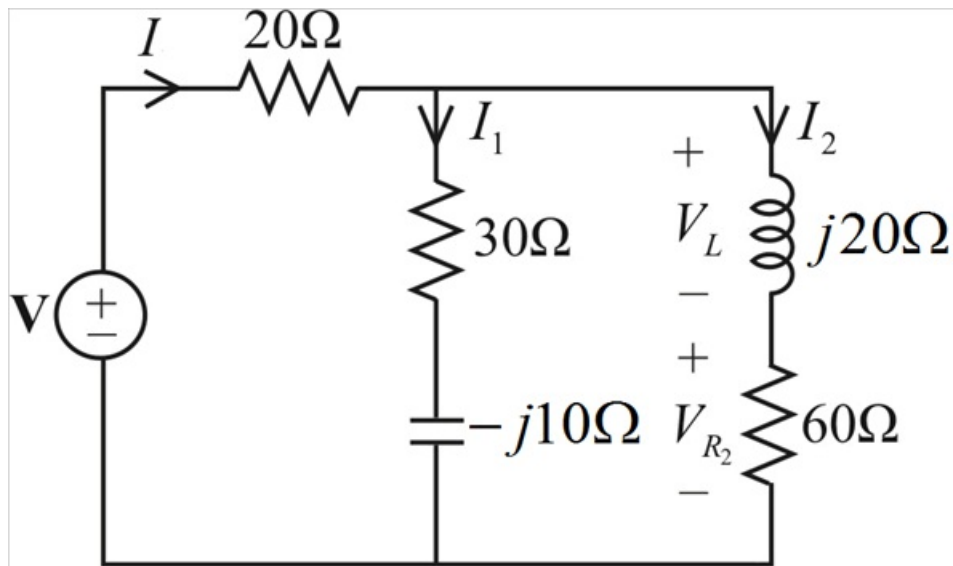
Figure 11.25:



Step-by-step solution

Step 1 of 14

The given circuit is



Step 2 of 14

Given average power absorbed by the $60\text{-}\Omega$ resistor is 240 W , then

$$P_{60\Omega} = V_{R_2} I_2$$

$$P_{60\Omega} = \frac{V_{R_2}^2}{60}$$

$$240 = \frac{V_{R_2}^2}{60}$$

Step 3 of 14

$$V_{R_2}^2 = 240 \times 60$$

$$\therefore V_{R_2} = 120 \text{ V}$$

$$I_2 = \frac{V_{R_2}}{60}$$

$$I_2 = \frac{120}{60}$$

$$\therefore I_2 = 2 \text{ A}$$

Step 4 of 14

The 60Ω and $j20 \Omega$ are in series, so their combined impedance is

$$60 \Omega + (j20) \Omega = 60 + j20 \Omega$$

The 30Ω and $-j10 \Omega$ are in series, so their combined impedance is

$$30 \Omega + (-j10) \Omega = 30 - j10 \Omega$$

The $(60 + j20) \Omega$ and $(30 - j10) \Omega$ are in parallel, so their equivalent impedance is

$$Z = (60 + j20) \parallel (30 - j10)$$

Step 5 of 14

$$Z = \frac{(60 + j20)(30 - j10)}{(60 + j20) + (30 - j10)}$$

$$Z = \frac{2000}{90 + j10}$$

$$Z = \frac{2000}{90 + j10} \times \frac{90 - j10}{90 - j10}$$

$$Z = \frac{180000 - j20000}{90^2 + 10^2}$$

$$Z = \frac{180000 - j20000}{8200}$$

$$Z = 21.951 - j2.439 \Omega$$

Step 6 of 14

By voltage division rule, voltage across the impedance Z is

$$V_Z = \frac{Z}{Z + 20} \text{ V}$$

$$V_Z = \left(\frac{21.951 - j2.439}{21.951 - j2.439 + 20} \right) \text{ V}$$

$$V_Z = \left(\frac{21.951 - j2.439}{41.951 - j2.439} \right) \text{ V}$$

$$V_Z = (0.5249 - j0.0276) \text{ V} \dots\dots\dots (1)$$

Step 7 of 14

Again by voltage division rule, voltage across 60Ω resistor is

$$V_{R_2} = \frac{60}{60 + j20} V_Z$$

$$V_{R_2} = \left(\frac{3}{3 + j1} \right) (0.5249 - j0.0276) \text{ V}$$

But $V_{R_2} = 120$

Step 8 of 14

$$120 = \frac{3}{3 + j1} \times \frac{3 - j1}{3 - j1} (0.5249 - j0.0276) \text{ V}$$

$$120 = \frac{9 - j3}{3^2 + 1^2} (0.5249 - j0.0276) \text{ V}$$

$$120 = \frac{9 - j3}{10} (0.5249 - j0.0276) \text{ V}$$

$$\text{V} = \frac{120}{(0.9 - j0.3)(0.5249 - j0.0276)}$$

$$\text{V} = \frac{120}{0.46413 - j0.18231}$$

$$\text{V} = 223.989 + j87.983 \text{ V}_{\text{rms}}$$

$$\therefore \boxed{\text{V} = 240.65 \angle 21.44^\circ \text{ V (rms)}}$$

Substituting V value in equation (1), we get $V_Z = (0.5249 - j0.0276)(223.989 + j87.983)$

$$V_Z = (120 + j40) \text{ V}$$

Step 9 of 14

Writing KVL in the left loop, we get

$$\text{V} = V_{20\Omega} + V_Z$$

$$V_{20\Omega} = \text{V} - V_Z$$

$$V_{20\Omega} = 223.989 + j87.983 - 120 - j40$$

$$V_{20\Omega} = 103.989 + j47.983 \text{ V}$$

$$\therefore V_{20\Omega} = 114.525 \angle 24.77^\circ \text{ V}$$

Step 10 of 14

Current through 20Ω resistor is,

$$I = \frac{V_{20\Omega}}{20}$$

$$I = \frac{103.989 + j47.983}{20}$$

$$\therefore I = 5.2 + j2.4 \text{ A}$$

Step 11 of 14

Thus, the complex power absorbed by the 20Ω resistor is

$$S_{20\Omega} = \frac{|V_{20\Omega}|^2}{20}$$

$$S_{20\Omega} = \frac{(114.525)^2}{20}$$

$$\therefore \boxed{S_{20\Omega} = 656 \text{ VA}}$$

Step 12 of 14

The complex power absorbed by the $(60 + j20)\Omega$ impedance is

$$S_{(60+j20)\Omega} = V_Z I_2^*$$

$$S_{(60+j20)\Omega} = (120 + j40) \times 2$$

$$\therefore \boxed{S_{(60+j20)\Omega} = (240 + j80) \text{ VA}}$$

Step 13 of 14

The complex power absorbed by the $(30 - j10)\Omega$ impedance is

$$S_{(30-j10)\Omega} = V_Z I_1^*$$

Writing KCL at center node, we get:

$$I = I_1 + I_2$$

$$I_1 = I - I_2$$

$$I_1 = 5.2 + j2.4 - 2$$

$$I_1 = 3.2 + j2.4 \text{ A}$$

Step 14 of 14

$$S_{(30-j10)\Omega} = (120 + j40)(3.2 - j2.4)$$

$$\therefore \boxed{S_{(30-j10)\Omega} = 480 - j160 \text{ VA}}$$

Thus, the overall complex power of the circuit is

$$S = S_{20\Omega} + S_{(30-j10)\Omega} + S_{(60+j20)\Omega}$$

$$S = 656 + (480 - j160) + (240 + j80)$$

$$\therefore \boxed{S = 1376 - j80 \text{ VA}}$$